

Generation and selection

Generate by role

One generation branch per column
Fit on TRAIN; select on VAL
Record any TRAIN+VAL refit
Record inputs used in generation
Apply schema, domain and observation rules

Full scientific artifact

255,540 rows
555 logical columns
Retained for internal analysis

Q6 candidate selection

Eligible roles
No relevant selection blockers
Active-window check
Complete selection manifest
Disclosed warnings

Selected candidate

23 logical columns
For diagnostic use
Warnings retained

Synthetic
data

Grades
and
blocker
flags

Evaluation and claims

Real evaluation reference

Record the split and evaluation scope

Q1-Q5: applicable checks

Q1 Marginal fidelity
Q2 Temporal coherence
Q3 Observability
Q4 Cross-modal consistency
Q5 Task and baseline diagnostics
Record assessment status

Assessment results

Evaluated: Pass / Warning / Fatal
Separate blocker flags

Assess each claim

Matching data and scope
Sufficient usable observations
All required checks pass or have permitted warnings
No relevant blockers
Evaluation history supports the claim
Warnings disclosed

Claim decision

All conditions met → Supported
Known failure → Withheld
Otherwise:
Too few usable observations or undefined estimate → Insufficient evidence
Missing or mismatched result → Not assessed